Sk Nishat Saama

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.07
Auxilium Convent School	Class XII (ISC)	2020	91.50%
Auxilium Convent School	Class X (ICSE)	2018	91.80%

EXPERIENCE

Research Assistant at Indian Institute of Technology, Hyderabad

NLIP Lab | BharatGen Project | Supervisor: Dr. Maunendra Sankar Desarkar

- Investigating cultural knowledge and reasoning in regional and frontier LLMs through culturally grounded datasets and benchmarks, with a focus on identifying gaps in their understanding of Indian contexts.
- Exploring alignment of LLMs with Indian cultural values, including mechanistic interpretability to understand cultural representations and potential interventions at the architectural and decoding levels.

PUBLICATIONS

PUSHPAK: Towards Probing Understanding of Sub-cultures through Cultural Heterogeneity, Practices, and Knowledge

Research Summary - Developed a scalable framework for evaluating community-level cultural understanding in LLMs, constructing a benchmark of 21K questions across 42 communities, 4 languages and 6 question types consisting of recall and reasoning tasks. Incorporated shared cultural practices and the migration of cultural knowledge across communities into the evaluation. Analyzed model behavior, performance gaps between task types, and multilingual consistency to characterize limitations in cultural grounding and cross-community reasoning.

Publication -First author | Manuscript under review at ACL ARR (To be submitted in NAACL 2027)

PROJECTS

Cultural Adaptation using LLMs [Github]

- Developed a framework to **adapt GSM8K** word problems to **Indian cultural contexts** using the DIWALI knowledge base while targeting preservation of mathematical meaning.
- Implemented and compared **direct prompting, cultural reference prompting, zero-shot chain-of-thought, semantic RAG, and iterative refinement** across multiple language model families and sizes.
- Extended the framework with **agentic RAG** for dynamic retrieval, drafting, verification, and feedback-driven revision.
- Built an evaluation pipeline assessing **mathematical equivalence, cultural authenticity, linguistic quality.**

Cultural Value Benchmarking and Alignment of LLMs [Github]

- Benchmarked **8 LLM configurations** against cultural profiles from **107 countries** using PCA-based Inglehart-Welzel cultural mapping from the World Value Survey(WVS).
- Curated a **4,800-example India-Australia cultural instruction corpus**, incorporating semantically diverse paraphrases generated using GPT-4o, Gemini, and Claude.
- **Fine-tuned** open-source LLMs using **LoRA and supervised fine-tuning** to improve cultural alignment and quantified shifts in their cultural-value representations.

AI Tutor Assessment | IndoML Datathon 2025 [Github]

- Engineered an end-to-end **DeBERTa-v3 NLP pipeline** across **2,476 labeled examples** to classify tutoring guidance, incorporating student dialogue, tutor identity, and response context.
- Optimized fine-tuning through **15 Optuna trials**, selective layer freezing, class-weighted loss, and mixed precision; implemented **three-phase Hard Example Mining** with **2×-3× weighting** for difficult samples.
- Evaluated **5 ensemble strategies** and hierarchical classification; achieved **0.5967 macro-F1 (+5.7%)** and **12.3% higher minority-class F1** using a weighted ensemble on internal evaluation.

ACHIEVEMENTS

Secured 8th position in IndoML Datathon (BERT based LLM Tutor Evaluation)

Secured 2nd position(pvt leaderboard) in KLA Hackathon(Image Restoration Task)

Secured 3rd position in MOTIVE Hackathon (RAG based Chatbot)

Shortlisted to represent IITH in Smart India Hackathon 2026 (top 30 among 200+ submissions)

POSITIONS OF RESPONSIBILITY

Teaching Assistant for **NLP, Topics in NLP,** IIT Hyderabad

Teaching Assistant for **Machine Learning -- INAE CEEE 2026**, IIT Hyderabad

Teaching Assistant for **Generative AI -- AIET Cohort 8 and 9**, IIT Hyderabad

SKILLS

- **Languages:** Python, C++, C, SQL, HTML, CSS, Shell Script
- **AI/ML Concepts:** Transformer Architectures, BERT, Large Language Models , Fine-Tuning, RAG, Agentic AI
- **Libraries & Tools:** PyTorch, Hugging Face, scikit-learn, NumPy, pandas, Matplotlib, CUDA, Neo4j, Git, Visual Studio
- **Interests:** NLP, GenAI, AI/ML, DL, Data Mining, Data Science, Data Analytics

RELEVANT COURSES

Natural Language Processing | Topics in NLP | Introduction to Program Analysis and Compiler Optimization | GPU Programming | Foundations of Machine Learning | Data Mining | Advanced Data Structures and Algorithms